CHAPTER II
Nyaya Epistemology

Introduction

The Vaishesika system is chronologically subsequent to Sähkhya and predates Nyäya. It is likely antecedent to, and absolutely not subsequent to, Buddhism and Jainism. The term originates from 'Vishesa', meaning particularity, a distinguishing trait, or distinctiveness. The Vaishesika philosophy is a pluralistic realism that underscores the notion that diversity constitutes the essence of the universe. This theory extensively examines the concept of Vishesa, or particularity, which is considered the essence of entities. The originator of this technique is Kanada, sometimes referred to as Kanabhuk, Ulüka, and Kâshyapa. This technique is often referred to as 'Kanada' or 'Aulüka Darshana'. He was named Kanada due to his lifestyle as an ascetic, subsisting on grains collected from the fields. 'Kana', while signifying 'grain', also denotes a particle or a particular, whereas the term 'Kanada' implies an individual who adheres to the philosophy of particularity—vishesa. Prashastapâda authored the renowned Padärthadharmasahgraha, referred to as a Bhäsya or Commentary on Kanada's Vaishesikasütra, yet it constitutes a highly valued standalone work. Udayana and Shridhara have commented on it. The Vaishesika was subsequently integrated with the Nyäya, which embraced the ontology of the former and elaborated upon it through its epistemological framework. Consequently, Shivädítya, Laugäksi Bhaskara, Vishvanätha, and Annambhatta address the two systems concurrently, analysing their interconnections and the implications of their combined philosophical perspectives on knowledge and reality.

The philosopher Gotama is the originator of the Nyaya School. He is also referred to as Gautama and Aksapada. Nyäya denotes arguments and implies that the system is primarily intellectual, analytical, logical, and epistemological. It is also referred to as Tarkashästra, the science of reasoning; Pramänashästra, the science of logic and epistemology; Hetuvidyä, the science of causes; Vädavidyä, the science of argument; and Ânvïksikï, the science of critical examination.

Nyāya is a framework of atomistic pluralism and logical realism. It is associated with the Vaishesika system, which is considered 'Samänatantra' or a comparable philosophy. Vaishesika

formulates metaphysics and ontology; Nyäya constructs logic and epistemology. Both concur in perceiving earthly existence as replete with misery, constituting a bondage of the soul, and in considering liberation, defined as the complete cessation of suffering, as the ultimate objective of life. Both concur that bondage arises from ignorance of reality, whereas release stems from accurate understanding of reality. Vaishesika addresses the elucidation of reality, while Nyäya focuses on the articulation of accurate understanding of reality. Nyāya predominantly endorses Vaishesika metaphysics. However, several significant distinctions between them warrant attention. Initially, the Vaishesika identifies seven categories and categorises all realities inside them, whereas the Nyäya delineates sixteen categories and encompasses all seven Vaishesika categories under one, termed 'Prameya' or 'the Knowable', which is the second category among the sixteen. The initial category is Pramäna, denoting the valid means of knowing. This distinctly highlights the primarily logical and epistemological nature of the Nyäya philosophy. Secondly, while the Vaishesika acknowledges solely two Pramänas—perception and inference—dismissing comparison and verbal authority as forms of inference, the Nyäya distinguishes all four as separate entities: perception, inference, comparison, and verbal authority. Knowledge (jnana) is defined as apprehension (upalabdhi) or consciousness (anubhava). Nyāya posits that knowledge elucidates both the subject and the object, which are fundamentally separate from it, thereby allowing individuals to understand the relationship between their perceptions and the external world. All knowledge is a revelation or manifestation of objects (arthaprakäsho buddhih). Similar to how a lamp illuminates tangible items positioned before it, knowledge unveils all entities that present themselves to it. Knowledge can be either valid or invalid. Valid knowledge (pramä) is defined as the accurate comprehension of an object (yathärthänu- bhavah). It is the representation of a thing in its true form. Nyäya upholds the connection theory. For knowledge to be valid, it must align with reality. Valid knowledge is generated by the four legitimate means of knowledge: perception, inference, comparison, and witness. Invalid knowledge encompasses recollection (smrti), doubt (samshaya), error (viparyaya), and hypothetical reasoning (tarka). Memory is invalid, as it constitutes representational cognition rather than presentative cognition. The object of remembrance is not immediately accessible to the soul but is only recalled indirectly. Doubt constitutes doubt in thought. Error arises from a misunderstanding, as it fails to align with the actual item. Hypothetical reasoning does not constitute genuine knowledge. It posits the argument, "If there is no fire, then there cannot be smoke." Recognising a rope as such demonstrates accurate understanding. If you are unsure if it is a rope or a snake, you may

harbour doubts. If you remember the rope you have observed, you possess memory. To confuse the rope with a snake constitutes an error.

Knowledge is generated within the soul through its interaction with the non-soul. This characteristic of the soul is incidental and is produced by the object interacting with it. If the conditions for generation are robust, knowledge is legitimate; if they are flawed, knowledge is invalid. A man with normal vision perceives a conch as white, whereas a man afflicted with jaundice perceives it as yellow. The essence of truth lies in its correspondence with the object. Knowledge is legitimate if it aligns with its object; if it does not, it is invalid. Valid knowledge aligns with its object (yathārtha and avisamvādi) and facilitates effective action (pravṛttisāmarthya). Erroneous knowledge fails to align with its object, resulting in failure and disappointment (pravrttivisamväda). Fire must burn, cook, and illuminate. If it fails to ignite, it is not a fire. Knowledge is fundamentally a manifestation of objects. The inquiry of its validity or invalidity is a subsequent matter that is contingent upon its alignment with its aim. Truth and falsehood are external attributes of knowledge. They are seized by subsequent understanding. They emerge and are understood only after knowledge has already manifested. They are neither inherent nor self-apparent. The validity and invalidity of knowledge emerge (utpattau paratah prämänyam) subsequent to the acquisition of knowledge, and they are recognised (jnaptau paratah prämänyam) following the emergence of knowledge and their occurrence. Correspondence constitutes the substance, whereas successful action serves as the criterion for truth, indicating that a belief is true if it corresponds to reality and leads to effective outcomes in practice. The Nyäya theory of knowing is thus both realistic in its understanding of nature and pragmatic in its criteria for truth. The four types of valid knowledge are perception, inference, comparison or analogy, and verbal witness.

Perception

Gotama characterises perception as 'accurate cognition arising from the interaction of the sense organs with objects, devoid of nomenclature, and distinctly delineated.' "This concept of perception omits divine and yogic perception, which is not produced by the interaction of the sense organs with external objects. Vishvanätha defines perception as 'direct or instantaneous cognition that is not obtained through the mediation of any other cognition. ' This concept encompasses both ordinary and extraordinary perception while excluding inference, comparison, and witness. Perception constitutes a form of knowledge and is an inherent characteristic of the self. Conventional perception assumes the presence of sense organs, objects, the mind, and the self, together with their interactions. The self interacts with the manas, the manas with the sensory organs, and the sensory organs with the objects. Contact

between the sense organs and things is contingent upon the manas first engaging with the sense organs, which in turn requires the self to interact with the manas. Therefore, interaction between sense objects inherently requires both contact between the manas and the senses, as well as contact between the self and the manas. The sensory organs originate from elements that exhibit unique attributes of smell, taste, colour, touch, and sound, which are essential for the process of perception and interaction with the environment. Perception constitutes a form of knowledge and is an inherent characteristic of the self. Conventional perception assumes the presence of sensory organs, objects, the mind, and the self, together with their interactions. The self interacts with the manas, the manas with the sensory organs, and the sensory organs with the objects. The interaction of the sense organs with objects is contingent upon the prior engagement of the manas with the sense organs, which in turn requires the self to first connect with the manas. Therefore, contact between sense objects inherently requires the interaction between the mind and the senses, as well as the interaction between the self and the mind. The sensory organs originate from elements that exhibit unique attributes for smell, taste, colour, touch, and sound. The manas serves as the intermediary between the self and the sensory organs, facilitating the processing and interpretation of sensory information to form perceptions and experiences. The external object, perceived through the senses and the manas, creates an impact on the self. Consequently, the theory is pragmatic.

The Naiyāyika asserts two phases in perception. The first is termed 'indeterminate' or 'nirvikalpa', whereas the second is referred to as 'determinate' or 'savikalpa'. They represent not two distinct forms of vision, but rather the initial and subsequent phases within the same intricate process of perception. Gotama acknowledges these two levels in his aforementioned description of perception. Perception is 'unassociated with a name' (avyapadeshya), signifying 'indeterminate', and it is 'well-defined' (vyavasayatmaka), indicating 'determinate'. All perception is definitive, but it is invariably preceded by a prior phase of indeterminacy, which allows for the initial gathering of sensory information before it is processed into a clear understanding. Nvâva acknowledges the essential principle about knowledge, attributed as Kant's unique contribution to Western philosophy, which asserts that knowledge encompasses both sense and idea. “Percepts devoid of conceptions are blind, whereas concepts devoid of percepts are vacuous.” Perception is a multifaceted process of experience that encompasses both sensation and cognition. All perception is determined, as it constitutes perceptual knowledge or judgement. Sensation constitutes the substance, while idea represents the structure of knowledge. Bare sensation or simple apprehension constitutes nirvikalpa perception, while perceptual judgement or relational apprehension represents savikalpa

perception. Nyāya circumvents the mistake of the psychical staircase hypothesis, which posits that one first has sensory experience, followed by conceptualisation and subsequently judgement. Perception is a multifaceted process of presentation and representation, wherein direct consciousness cannot be distinctly separated from relational judgement. Indeterminate perception constitutes the substance from which determinate perception is formed; yet, they can only be differentiated conceptually and cannot be separated in actuality. Nirvikalpa perception is the initial apprehension, pure consciousness, and direct sensory experience that is undifferentiated, non-relational, and devoid of assimilation, discrimination, analysis, and synthesis. The awareness of "that" is not yet defined by the awareness of "what". However, since the 'that' cannot be comprehended independently from the 'what', and the 'substance' cannot be understood apart from its 'qualities', we promptly arrive at savikalpa perception. Here, the mere awareness of the 'that' and the 'what', along with their 'inherence' as an undifferentiated, unrelated, mute, and inarticulate entity, is transmuted into differentiated, relational, conceptual, and articulate knowledge that encompasses assimilation, discrimination, analysis, and synthesis. For instance, upon entering a dark cinema hall from bright daylight to attend a matinée, our initial view of the seats and audience is obscured, presenting only a vague awareness of the objects, which progressively become discernible; this indistinct perception of the objects in the hall constitutes indeterminate perception, whereas the subsequent clear recognition of them represents determinate perception. The simple recognition of an object as a 'that' is indeterminate perception, but the distinct perception of it along with its characteristics represents determinate perception. At dusk, we observe a linear object on the road and ascertain upon closer inspection that it is a rope. A distant white object approaches, revealing itself to be a white cow. The initial phase is ambiguous, while the subsequent phase is characterised by determined perception. We are pressed for time and wish to complete our bath before departing. It is uncertain whether the water was cold and the bath invigorating; however, we did perceive the water's coolness and the bath's refreshing nature. We perceive water and its coolness, although we do not associate the two sensations. Indeterminate perception reveals the thing in its unadorned state, devoid of any characteristics. In determinate perception, we associate the substance with its properties. The sensation of ambiguous perception is psychological, while its understanding is logical. In its purest form, as just awareness, we perceive an indeterminate sensation; we experience it, yet the instant we attempt to comprehend it as 'bare awareness', it transitions into conception and becomes determinate. Therefore, all our perceptions, as forms of cognition, are determinate and constitute perceptual judgements. Indeterminate perception may only be conceptually distinguished from determinate perception, not in actuality, which

means that in practice, our experiences of indeterminate perception are often influenced by prior knowledge and contextual factors that shape our interpretations. Therefore, while we perceive ambiguous perception as a psychological state of sensory experience, our understanding of it, even as indeterminate perception, arises from logical deduction. We experience it directly, although it is just awareness, not cognition. Pure apprehension, being an infra-relational phenomenon, cannot be comprehended without the context of cognitive processes that follow, which transform it into meaningful experiences. Cognition is inferred thereafter when conceptualisation has converted a mere sensation into a perceptual judgement.

Vātsyāyana asserts that when an item is recognised by its name, it results in determinate perception; conversely, when observed without its name, it leads to indeterminate perception. Jayanta Bhatta asserts that indeterminate perception recognises material, characteristics, and acts, as well as universals, as independent and indistinct entities without any association with a name, whereas determinate perception comprehends all these elements in conjunction with a name. Gangesha Upädhyäya characterises indeterminate perception as the non-relational recognition of an item free from any association with a name, genus, or differentia. Annam Bhatta defines indeterminate perception as the direct perception of an item and its attributes, without understanding the relationship between the two. The 'that' and the 'what' are seen separately, and it is not understood that the 'what' characterises the 'that' or that these qualities inhere in the substance. 'Indeterminate perception' refers to 'mere acquaintance', which William James describes as 'raw, unverbalised experience', whereas 'determinate perception' denotes relational apprehension. Perception can be classified as either ordinary (laukika) or extraordinary (alau-kika). Laukika perception occurs when the sense organs interact with objects in a conventional manner. If the interaction between the sense organs and things occurs in an atypical manner, meaning the items are not normally accessible to the senses but are transmitted through an extraordinary channel, we experience Alaukika perception. Perception is categorised into two types: internal (mänasa) and outward (bähya). The mind (manas), as the internal organ, engages with psychical states and processes like cognition, emotion, volition, desire, pain, pleasure, and aversion. External perception occurs when the five sensory organs interact with external objects. There are five types: visual, auditory, tactile, gustatory, and olfactory, which are elicited by the sensory organs of sight, sound, touch, taste, and smell, respectively, upon interaction with external objects. The external sense organs consist of the material elements of earth, water, fire, air, and ether, and so each perceives the specific quality

of its corresponding element. Consequently, the olfactory sense organ consists of terrestrial atoms and detects odours, which represent the distinctive quality of earth, and so on.

Extraordinary perception is categorised into three types: sämänyalaksana, jnänalaksana, and yogaja. Sämänyalaksana perception refers to the recognition of universals. Nyaya posits that universals constitute a separate category of realities. They reside in the specifics associated with various categories due to the distinct universals inherent in them. An individual is classified as belonging to a specific category due to the inherent qualities of that category residing inside it. Consequently, a cow is classified as a cow due to the inherent quality of universal cowness within it. Typically, we recognise only the specifics and not the generalities. We recognise specific cows, but we do not sense a 'universal cow'. Thus, the Nyäya asserts that universals are sensed in an exceptional manner. When we observe a specific cow, we initially recognise the inherent 'universal cowness' inside it. The second type of unusual perception is referred to as jnänalaksana perception. It is the intricate perception derived from association. Occasionally, disparate sensations get linked, resulting in a unified perception, which can lead to a more complex understanding of an experience or concept. An item is not directly viewed by a sense organ; rather, it is recalled from memory through prior cognition and is apprehended via representation. For instance, I observe a blossoming rose from afar and state, ‘I perceive a fragrant rose. 'However, how can scent exist? It can just be perceived through smell. Fragrance is detectable via the olfactory sense, whereas the visual sense is limited to perceiving colour. The visual perception of the rose evokes the recollection of its aroma through associative recall, previously experienced through olfactory senses. The experience of the aromatic rose through sight is referred to as 'jnänalaksana perception', or the recollection of the item based on prior cognition (jnäna). Additional instances include 'the piece of sandalwood appears aromatic’, ‘ice appears frigid’, ‘stone appears solid’, ‘tea appears steaming’, etc. The Nyäya theory of illusion, termed 'Anyathäkhyäti', is founded on this type of perception. When we confuse a rope with a snake, the concept of the snake, previously seen, is recalled by this remarkable jnänalaksana perception and is conflated with the object (i.e., rope) that is immediately perceived by the sense organ. When a shell is erroneously identified as silver, the memory of silver previously encountered at a shop (äpanastha) or elsewhere is reactivated by jnanalaksana perception and conflated with the item (i.e., shell) that is directly received by the sense organ. The past impression signifies the object in our consciousness. The error arises from an incorrect synthesis of the presented and represented objects. The depicted object is mistaken for the displayed one. The term 'anyathä' signifies 'otherwise' and 'elsewhere', and both meanings are

manifested in a mistaken perception. The exhibited object is interpreted differently, while the represented thing exists in a different location. The shell, the silver, the rope, and the snake all exist independently; their combination is merely an illusion. The shell and the rope are explicitly identified as 'this' (when we state: 'this is silver' or 'this is a snake'), whereas the silver and the snake reside in a different realm and are recalled through jnänalaksana perception. The third type of exceptional perception is termed 'yogaja perception'. The Yogins gain an intuitive and immediate perception of all objects—past, present, and future—through the power of meditation. It resembles the Kevalajnana of the Jains, the Bodhi of the Buddhists, the Kaivalya of the Särtkhya-Yoga, and the Aparoksanubhuti of the Vedantins. It is instinctive, transcendent, and beyond relational dynamics.

Inference

The second type of knowledge is anumä, which is inferential or relational, and its method is referred to as anumäna, or inference. It is characterised as a cognition that presupposes another cognition. It is mediated and indirect, arising through a 'mark', the 'middle term' (linga or hetu), which is inevitably associated with the 'main term' (sädhya). It is knowledge (mäna) that emerges after (anu) other knowledge. Invariable concomitance (vyâpti) is the essence of inference. The inclusion of the middle term within the minor term is referred to as 'paksadharmatä'. The constant correlation of the middle term with the major phrase is referred to as 'vyâpti'. The understanding of paksadharmatä, as defined by vyâpti, is referred to as parämarsha. Inference is described as knowledge derived from parämarsha, namely the awareness of the major in the minor through the intermediary that exists inside the minor (paksa-dharmatä) and is consistently linked to the major (vyäpti). Similar to the Aristotelian syllogism, Indian inference comprises three terms. The major, minor, and middle are referred to as 'sädhya', 'paksa', and 'linga' or 'hetu', respectively. Smoke is consistently linked to fire (vyäpti), and upon observing smoke on a hill, we deduce the presence of fire in that location. Hill constitutes the minor term; fire represents the major phrase; smoke serves as the middle term. Based on the observation of smoke on the hill and the principle that smoke indicates the existence of fire, we deduce that fire exists on the hill. This constitutes inference. Indian logic does not distinguish between reduction and induction. Inference is a multifaceted

process that encompasses both elements. Indian logic also refutes the verbalist perspective on logic. It examines cognition in its entirety rather than merely its manifestations. The formal and material logic are integrated here. The verbal form does not constitute an essential component of the inference. This is evident from the classification of inference into svärtha (for oneself) and parärtha (for others). In the former, a formal explanation of the various components of inference is unnecessary. It constitutes a psychological process. The parärtha, which is a syllogism, must be articulated in language solely to persuade others. The Nyāya syllogism comprises five components. The initial term is referred to as 'Pratijnä' or 'proposal'. This is the logical proposition that requires proof. The second is Hetu, or 'reason', which articulates the rationale for the establishment of the proposition. The third is termed Udäharana, which provides universal concomitance along with an illustration. The fourth is Upanaya, referring to the application of universal concomitance to the current issue. The fifth is Nigamana, or the conclusion derived from the prior assertions. The five assertions of the Indian syllogism are referred to as 'members' or 'avayavas'. A standard Nyāya syllogism is presented as follows:

This hill possesses fire (pratijnä).

(2) Due to the presence of smoke (hetu).

(3) Wherever there is smoke, there is fire; for instance, an oven (udäharana).

This hill emits smoke that is consistently linked to fire (upanaya).

Consequently, this hill possesses fire (nigamana).

When compared to the Aristotelian syllogism, which consists of three propositions, the Nyāya syllogism aligns with the Barbara (AAA) mood of the first figure, recognised as the most robust mood of the strongest figure. Although the Nyäya syllogism comprises five assertions and the Aristotelian syllogism consists of three, both use only three terms: sädhya (major), paksa (minor), and hetu (middle). Among the five propositions, two seem superfluous, allowing us to omit either the initial two or the last two, as they are fundamentally identical. The first aligns with the fifth, and the second corresponds with the fourth. Excluding the last two, the initial three propositions align with the conclusion, the minor premise, and the major premise, respectively. If we exclude the first two, the final three assertions align with the major premise, the minor premise, and the conclusion of the Aristotelian syllogism. Consequently, if we

exclude the initial two components of the Nyāya syllogism, which are encompassed inside the final two, we observe that it bears resemblance to the Aristotelian syllogism. First Figure:

(1) All things which have smoke have fire (major premise).

(2) This hill has smoke (minor premise).

(3) Therefore this hill has fire (Conclusion).

And the typical Aristotelian syllogism may be stated in the Nyäya form thus:

(1) Socrates is mortal (pratijnä).

(2) Because he is a man (hetu).

(3) Whoever is a man is a mortal, e.g., Pythagoras (udäharana).

(4) Socrates is a man who is invariably a mortal (upanaya).

(5) Therefore Socrates is mortal (nigamana).

However, there exist distinct discrepancies between the Nyäya and Aristotelian syllogisms, beyond the mere notional variation in the number of propositions in each. The Aristotelian syllogism is only deductive and formal, whereas the Nyäya syllogism is both deductive-inductive and formal-material. The Nyäya appropriately perceives deduction and induction as intrinsically connected, representing two facets of the same process—an understanding now acknowledged in Western logic. Inference, as per Nyäya, is derived not from the universal to the particular or vice versa, but rather from the specific to the particular via the universal. This example exemplifies a distinctive characteristic of the Nyäya syllogism, demonstrating that the universal major premise arises from genuine induction grounded in the rule of causation and that induction and deduction are inherently interconnected. In the Aristotelian syllogism, the main and minor words are distinct in the premises; however, they are linked by the middle term. Conversely, in the Nyäya syllogism, all three elements are integrated inside the Upanaya. While the Aristotelian syllogism is primarily verbal, the Nyäya acknowledges that verbal expression is not the substance of inference and is merely necessary for persuading others. There exist distinct basic disparities between the two perspectives. We more reasonably attribute the similarities between the two to a parallel evolution of thought. Indian logic has developed organically.

There are five characteristics of the middle term:

(1) It must be present in the minor term (paksadharmatâ); e.g., smoke must be present in the hill.
(2) It must be present in all positive instances in which the major term is present; e.g., smoke must be present in the kitchen where fire exists (sapaksasattva).
(3) It must be absent in all negative instances in which the major term is absent; e.g., smoke must be absent in the lake in which fire does not exist (vipaksäsattva).
(4) It must be non-incompatible with the minor term; e.g., it must not prove the coolness of fire (abädhita).
(5) It must be qualified by the absence of counteracting reasons which lead to a contradictory conclusion; e.g., ‘the fact of being caused’ should not be used to prove the ‘eternality of sound' (aviruddha).

Inference is typically classified into two types: Svärtha and Parärtha, which we have previously examined. Gotama delineates three categories of inference: pûrvavat, shesavat, and sämänyatodrsta. The initial two are predicated on causation, whereas the final one is founded on mere coexistence. A cause is the constant and unconditional precursor of an effect, while an effect is the constant and unconditional result of a cause. Pûrvavat inference occurs when we deduce an unobserved result from a visible cause, such as inferring impending rain from the presence of dark clouds in the sky. When we deduce an unobserved cause from a perceived consequence, we engage in shesavat inference; for instance, inferring past rainfall from the rapidly flowing, muddy, and inundated water of a river. Inference derived from regularity of co-existence rather than causality is termed 'sämänyatodrsta'; for instance, inferring the presence of cloven hooves in an animal based on its horns. One interpretation posits that a pûrvavat inference relies on prior experience of universal concomitance between two entities, a shesavat inference is characterised as 'parishesa' or inference by elimination, and a sämänyatodrsta is defined as inference by analogy.

Another classification of inference includes kevalänvayi, kevala vyatireki, and anvaya vyatireki inferences. It is based on the essence of vyäpti and the many methodologies for its establishment. The methods of induction that demonstrate universal causal relationships may include anvaya, vyatireka, or a combination of both. The first pertains to Mill's Method of Agreement, the second to his Method of Difference, and the third to his Joint Method of Agreement and Difference, often known as the Method of Double Agreement. Kevalänvayi inference occurs when the middle term is consistently positively correlated with the major

word. The concepts concur solely in presence, with no negative instance of their concurrence in absence, e.g.,

All knowable objects are nameable;

The pot is a knowable object;

The pot is nameable.

We have kevalavyatireki inference when the middle term is the differentium of the minor term and is always negatively related to the major term. The terms agree only in absence, there being no positive instance of their agreement in presence, e.g.,

What is not different from other elements has no smell;

The earth has smell;

The earth is different-from-other-elements.

We have anvayavyatireki inference when the middle term is both positively and negatively related to the major term. The vyäpti between the middle and the major is in respect of both presence and absence. There is double agreement between the terms—they agree in presence in the positive instances, and they also agree in absence in the negative instances; e.g.,

All things which have smoke have fire;

This hill has smoke;

.'. This hill has fire; and

No non-fiery things have smoke;

This hill has smoke;. This hill is not non-fiery; i.e., This hill has fire.

In Indian logic a fallacy is called 'hetvabhasa'. It means that the middle term appears to be a reason but is not a valid one. All fallacies are material fallacies. We have mentioned the five characteristics of a valid middle term. When we violate these characteristics, we encounter fallacies. Five kinds of fallacies are recognised:

(i)Asiddha or Sädhyasama: This is the fallacy of the unproved middle. The middle term must be present in the minor term (paksadharmatä). If it is not, it is unproved. It is of three kinds—

(a) äshrayä siddha: The minor term is the locus of the middle term. If the minor term is unreal, the middle term cannot be present in it; e.g., 'the sky-lotus is fragrant, because it is a lotus, like the lotus of a lake'.

(b) svarüpä siddha: Here the minor term is not unreal. But the middle term cannot by its very nature be present in the minor term; e.g., 'sound is a quality because it is visible'. Here visibility

cannot belong to sound which is audible

(c) vyäpyatvä siddha: Here vyâpti is conditional (sopädhika). We cannot say, e.g., 'Wherever there is fire, there is smoke.' Fire smokes only when it is associated with wet fuel. A red-hot iron ball or clear fire does not smoke. Hence, 'association with wet fuel' is a condition necessary to the aforesaid vyâpti. Being conditioned, the middle term becomes fallacious if we say, 'The hill has smoke because it has fire.'

(2) Savyabhichära or Anaikäntika: This is the fallacy of the irregular middle. It is of three kinds.

: (a) Sädharana: Here the middle term is too wide. It is present in both the sapaksa (positive) and the vipaksa (negative) instances and violates the rule that the middle should not be present in the negative instances (vipak-säsattva); e.g., 'the hill has fire because it is knowable'. Here 'knowable' is present in fiery as well as nonfiery objects.

(b) Asädhärana: Here the middle term is too narrow. It is present only in the paksa and neither in the sapaksa nor in the vipaksa. It violates the rule that the middle term should be present in the sapaksa (sapaksasattva); e.g., 'sound is eternal because it is audible. Here, audibility belongs to sound alone and is present nowhere else.

(c) Anupasarhhäri: Here the middle term is non-exclusive. The minor term is all-inclusive and leaves nothing by way of sapaksa or vipaksa; e.g., 'All things are noneternal because they are knowable.'

(3) Satpratipakça: Here the middle term is contradicted by another middle term. The reason is counterbalanced by another reason. And both are of equal force; e.g., 'sound is eternal because it is audible' and 'sound is non-eternal because it is produced'. Here 'audible' is counterbalanced by 'produced', and both are of equal force.

(4) Bâdhita: It is the non-inferentially contradicted middle. Here the middle term is contradicted by some other pramäna and not by inference. It cannot prove the major term, which is disproved by another stronger source of valid knowledge; e.g., 'fire is cold because it is a substance. Here the middle term 'substance' becomes contradicted because its major term 'coldness' is directly contradicted by perception.

(5) Viruddha: It is the contradictory middle. The absence of the major term pervades the middle term, not its presence. Instead of proving the existence of the major term in the minor term, it proves its non-existence therein; e.g., 'sound is eternal because it is produced.' Here 'produced', instead of proving the eternality of sound, proves its non-eternality. Here the middle term itself disproves the original proposition and proves its contradictory, while in the savyabhichära the middle term only fails to prove the conclusion, and in the satpratipaksa the middle term is inferentially contradicted by another middle term, both of which are of equal force, and in the bâdhita the middle term is non-inferentially contradicted and the major is disproved by a stronger pramäna other than inference.

Comparison

The third type of legitimate cognition is called Upamiti, and the corresponding means of obtaining it is referred to as Upamâna. This type of cognition involves information that is obtained through comparison and is approximately aligned with analogy. It is described as understanding the relationship between a word and its denotation. It is generated by the understanding of likeness or similarity. A guy who is unfamiliar with a gavaya, or wild cow, learns that a wild cow resembles a cow and later encounters one in a jungle, identifying it as such; his understanding is attributed to upamâna. He has encountered the term 'gavaya' and has been informed that it resembles a cow; now he perceives the thing signified by 'gavaya' and acknowledges its resemblance. Thus, upamâna refers to the understanding of the relationship between a name and the object it signifies. The recognition of similarity produces this understanding, as an individual identifies a wild cow as a 'gavaya' upon perceiving its resemblance to a cow and recalling the description that defines a 'gavaya' as an animal akin to a cow. Buddhists construe Upamäna as perception and testimony. The Sänkhya and Vaishesika schools limit it to inference. The Jainas simplify it to recognition or pratyabhijñā. The Mimâmsakas acknowledge it as an independent source of knowledge; nonetheless, their interpretation diverges from that of Nyäya, which will be examined in the chapter on Mimämsä.

Verbal Testimony

The fourth category of genuine knowledge is Shabda, Agama, or authoritative verbal testimony. Its significance is often referred to as Shabda. It is characterised as the assertion of a reliable individual (aptavakya) and entails comprehending its significance. A phrase is characterised as a compilation of words, whereas a word is described as an entity capable of conveying meaning. The ability of a word to express its meaning originates, as per old Nyāya, from God, and according to later Nyāya, from long-standing convention. Testimony is inherently subjective. It is founded on the statements of a reliable individual, whether human or divine. Testimony is categorised into two types: Vaidika and laukika (secular). The Vaidika witness is flawless and infallible, as the Vedas are divinely articulated; in contrast, secular testimony, being articulated by flawed humans, lacks infallibility. Only the statements of reliable individuals who consistently convey the truth hold validity; those of others do not. A word is a powerful symbol representing an object, while a sentence is a compilation of words. A phrase must adhere to specific requirements to be comprehensible. There are four conditions: äkänksä, yogyatä, sannidhi, and tätparya. The initial concept is mutual implication or anticipation. The words in a phrase are interconnected and depend on each other to convey a complete meaning. An arbitrary collection of disjointed words will not constitute a coherent sentence. It will be utter nonsense, for example, 'cow horse man elephant'. The second criterion is that the language must be appropriate to communicate the intended meaning and must not contradict it. "Watering the plants with fire is a paradoxical statement." The third criterion is the close proximity of the words to each other. The words must be articulated in rapid succession without prolonged pauses. If the words 'bring', 'a', and 'cow' are articulated at extended intervals, they would not constitute a coherent phrase. The fourth requirement pertains to the speaker's intention when the words are uncertain. The term 'saindhava' signifies both 'salt' and 'horse'. When a man consuming food requests another to bring 'saindhava', the latter should not present a horse.

The Nyäya recognises solely these four pramänas. Arthāpatti, or implication, is distilled to inference. For instance, when we state, 'Fat Devadatta does not consume food during the day,' it implies that he must be eating at night; otherwise, how could he maintain his weight? Mïmâmsâ confers the status of an autonomous pramāṇa to implication. However, Nyäya simplifies it to inference as follows:

All fat persons who do not eat during the day eat during the night.
Devadatta is a fat person who does not eat during day;
Devadatta is a fat person who eats during the night.

Abhâva, or non-existence, which also is regarded as a separate pramäna by Bhâtta Mïmâmsâ, is reduced here either to perception or to inference. Abhâva is the non-existence of a thing, and the same sense organ which perceives a thing also perceives its non-existence. If the thing is imperceptible and can only be inferred, then its non-existence too may be equally inferred.

Nyaya epistemology makes a substantial contribution to Indian philosophy by providing a systematic and logical framework for the attainment of genuine knowledge. The Nyaya school emphasises the importance of rational inquiry, critical thinking, and logical analysis in the pursuit of truth. It clarifies that genuine knowledge (prama) is obtained from reliable sources known as pramanas. Nyaya epistemology clarifies the sources of valid knowledge and promotes a rational and analytical method of reasoning. The emphasis on logic, evidence, and critical reasoning has significantly influenced Indian philosophical traditions and is vital for understanding the essence and attainment of knowledge.

In the fifth generation of computers, considerable effort has been dedicated to simulating intelligent activity in machines. Researchers are endeavouring to address this issue by employing grammar-based computing, which aims to impart Knowledge of Language (KOL) to machines for Natural Language Understanding (NLU) or Processing (NLP), alongside knowledge-based computing, which seeks to represent background/contextual or Real-World Knowledge (RWK) in machines for comprehending real-world scenarios and problem-solving. The researchers are contemplating both options. A growing number of individuals are recognising that RWK representation is essential for the simulation of consciousness in machines. No comprehensive framework has been produced that is applicable to all forms of

knowledge, identities of the location, object, and instrument of cognition. Numerous scholars in the domain of artificial intelligence are starting to interrogate the veracity of the following assertions:

* Knowledge can be represented
* Computers can encode and decode knowledge in everyday language
* Computers are capable of thinking and understanding
* Computers require knowledge in order to communicate with people or other machines

While Panini's grammar is indeed relevant to RWK, it mostly serves as a framework for KOL that can be utilised as a model for more languages. This is due to the unequivocal accounts of both the KOL and the RWK in India, especially in the Sanskrit language. One can locate an early account of RWK within the Naya system, which bears a resemblance to the views presented by AL philosophers. Consequently, it is imperative to examine the offerings of the Naya system. The objective is to compare and contrast the Indian epistemic tradition, particularly the Naya, with those linked with artificial intelligence.

There are two major fields of research in artificial intelligence: knowledge representation (KR) and information retrieval (IR). Knowledge can be characterised as objects and attributes, or even as an algorithm that contains objects and procedures, when approached from the perspective of artificial intelligence (AI). An interaction between "data" and "processes" or "abilities" and "memory storage" is stated to be the cause of the so-called "intelligent behaviour" that is observed. According to Jean-Raymond Abrial (1974), a database is a representation of the physical world as it is always changing. During a particular moment, the state of this model is a representation of the knowledge that it has obtained from the outside world. But knowledge is more than static encoding of facts in interacting with the world. The idea that knowledge of anything is the capacity to construct a mental model that accurately depicts the thing in question, as well as the actions that may be carried out by it and on it, is one of the fundamental premises of artificial intelligence (AI). Following this, individuals (or robots) are able to make predictions about what is likely to occur in the real world by testing actions on the model (Sowa in 1984). An earlier challenge faced by those working on artificial intelligence was determining whether to represent knowledge by using a declarative approach or a procedural method. For the purpose of knowledge representation, it has been finally

decided that the most appropriate strategy would be one that combines a declarative approach with a procedural approach. Predicate calculus, semantic nets, frames, conceptual graphs, scripts, semantic primitives, and neural nets are some of the other well-known techniques for KR. These techniques follow a wide variety of strategies for KR. In spite of this, reducing the unknown phenomenon known as knowledge to a single or two definitions would not be an aim in and of itself. According to Hopcroft and Lipman (1979), not only would it be necessary to comprehend the entirety of the human cognitive process, but it would also be necessary to develop efficient methods for representing information.

pratyaksa (perception)

* anumana (inference)

* upamana (analogy)

* sabda (verbal testimony) and latter is sub-classified as

* samsaya (doubt)

* viparyaya (error)

* tarka (hypothetical argument)

* smrti (memory)

5.1. **Comparison with AI**

The following sub-sections define each of the above sub-classifications of invalid and valid knowledge and give their AI parallels--

5.1.1. apramana--the invalid knowledge

It is defined as false knowledge, doubtful or illusionary. The four types of invalid knowledge are

defined and compared with AI as follows:

5.1.1.1. smrti (memory)

* is impressions left by previous knowledge

* it is non valid because the object of knowledge is absent at the time of remembrance

* AI explanation

* memory as gaps in knowledge or probabilistic knowledge which must be corrected before processing starts and conclusion is arrived at

* such gaps or incomplete knowledge may be replaced by new knowledge [right arrow] nonmonotonic knowledge (McDermott & Doyle 1980)

* static memory [right arrow] knowledge of object (WHAT)

* short term : memory in current focus

* long term: memory for future use

* procedural memory [right arrow] knowledge of behavior (HOW)

* short term

* long term

5.1.1.2. samsaya (d * conflicting judgment on the precise character of an object of cognition (Bijalwan 1987)

* neither true or false, only wavering judgment

* AI explanation

* ambiguity resolution and bug handling come close to samsya

5.1.1.3. tarka (hypothetical judgment)

* AI explanation

* Heuristics

* when reality is not known, truth is ascertained trying alternatives

* atmasraya (petito principi)

* if A [right arrow] A then A must be different from A

* anyonyasraya (dependent tarka)

* A[right arrow]B[right arrow]A

* cakraka (circular logic)

* recursion in AI

* A[right arrow]B[right arrow]C[right arrow]A

* anavastha (regressus ad infinitum or transitivity)

* A[right arrow]B[right arrow]C[right arrow]D ...

* tadanya-budhiat-prasanga (reduction ad absurdum)

* proves a proposition by negating the contradiction of the proposition

* A[right arrow]-(-A) or + [right arrow] -(-)oubt)

5.1.1.4. viparyaya (error)

* manifestation of a real object in another * positive misconception, incomplete knowledge, gaps in knowledge * AI explanation * can be compared to bugs and inconsistencies

5.1.2. pramana--the valid knowledge

The following section discusses the means to valid knowledge and compares them with AI

5.1.2.1. **pratyaksa (perception)**

* Five senses--sight, smell, taste, hearing, touch (mind [right arrow] sixth sense for cognition of
feelings like pleasure, pain etc)
* six ways of perception for six kinds of objects
* substance, quality, universal quality, sound, sound-ness, nonexistence
* AI [right arrow] icon, schema and percept
* icon [right arrow] temporary record of a sensory input that the brain keeps
* schema [right arrow] blue print of the mental model
* percept [right arrow] assembling units which are to be assembled according to the schema to
create a mental model
* Sowa (1984) calls the search mechanism "associative comparator" which must have the following
characteristics:
* associative retrieval
* brain is like a computer, retrieves data by an address in the storage by matching the best pattern
* top-down match
* perception searches for percepts that match the overall pattern of an icon
* stimulus constancy
* stimuli from same external object are recognized as equivalent despite varying size, brightness etc
* distributed storage
* images are not stored at a specific point, but distributed

5**.1.2.2. anumana (inference)**

* Gautama presupposes perception
* Jayanta gives five processes of inferring
* Perception of reason
* Remembrance of the universal concomitance
* Judgment that the subject of inference contains sense which, is concomitant with the object
* Knowledge of the consequence, and

* Judgment that the consequence is worthy of being accepted or rejected

* AI

* inference is very important and strong point of predicate calculus

* axioms [right arrow] theorems

* new facts can be deduced from axioms using rules of inference similar to ones given by Gautama

Gautama gives 3 ways to infer:

* psrvavat (deduction)

* cause [right arrow] effect.

* cloud [right arrow] rain

* "Gauri eats grass" from "Gauri is a cow" and "all cows eat grass" --

(inst gauri cow)

(forall X (if (inst X cow) (food X grass)))

(food gauri grass)

* sesavat (abduction)

* process [right arrow] explanations

* effect [right arrow] cause

* river is swollen [right arrow] there could have been rain

* Abduction has the following paradigms

From: b

(if a b)

Infer: a

* unlike deduction, abduction is not a legal interference. It can lead to false conclusions--

from: (feels nervous Ira)

(forall (X) (if (is sick X)

(feels nervous X)))

infer: (is sick Ira)

* samanyatod (induction)

* infers consequent from the antecedent, which is neither cause nor effect

* while induction can take several forms, the most common is

from: (P a), (P b),...

infer: (forall (X) (P X))

* although this is not a sound inference, induction like abduction, is very useful in everyday life where
it is more commonly known as learning
* for example, if we see a lot of leaves of green color, we might infer that all leaves are of green color
from: (if(inst leaf-1 leaf) (colour leaf_1 green))
(if(inst leaf-2 leaf) (colour leaf_2 green))
Infer: (forall (X) (if(inst X leaf) (colour X green)))
* Vacaspati divides inference into vita and avita (Bijalwan 1987). The former is based on universal
agreement in presence. As an example, we can infer
from: whatever has smoke has fire the hill has smoke
infer: the hill has fire
* avita is based on universal agreement in absence
from: whatever is non-different from other objects has no
smell. Earth has smell
infer: Earth is different from others.
* Prasastapada has svarthanumana and pararthanumana as two types of inference.
* Svarthanumana
* the premises are known from our own experience
*pararthanumana
* They are discovered by one man and imparted into another through language.
* Therefore, there is greater chance of error in pararthanumana which is based on words.svarthanumana is a mental process which is divided into data and samanyatodata by Prasastapada(Bijalwan 1987).
* Udyotakara gives kevalanvayin (K), kevalavyatirekin (KV), and amayavyatirekin (AV) as three types of inference. If a middle term is only positively related to the major term it is called K. Here. Thereason exists in the subject and similar instances, and is devoid of dissimilar instances. For example--we can infer "pot is nameable" from "pot is knowable" and "all knowable things are nameable"--
(property X nameable)))
(if (inst pot knowable)

(property pot nameable))

(property pot nameable)

In the case of KV, the middle term is negatively related to the major term. Reason exists in the subject, but not in dissimilar instances. For example--

from: (no (not P) is (M))

S is M

infer: S is P

When the middle is positively as well as negatively related to the major term, and reason exists in the entire subject, and in all similar instances but does not exist in dissimilar instances, the case is that of AV, as "sound is non-eternal because it is produced like a jar". The inference of Fire from smoke is also of this kind. Jayanta does not however, accept this division of inference in to K, KVand AV by Udyotakara (Bijalwan 1987).

5.1.2.3. upamana (analogy)

* upamana is the means by which we gain the knowledge of a previously unknown object on thebasis of its similarity to another object previously well known

* Mimansists and the Vedantists also accept analogy as an independent source of knowledge

* AI

* analogy is too slippery to be seen as just another road to induction

* often, analogy matches are so slipshod that they merely suggest solutions to the problem ofknowledge acquisition in general, and for inductive learning in particular

* analogy is useful because it serves as initial framework for developing the new concepts

5.1.2.4. sabda (verbal testimony)

* Leaving aside the Carvakas, Vaisesikas and the Buddhists, all the other systems of Indian philosophy accept sabda as a distinct source of knowledge, but they also differ with one another with respectto its nature forms and a number of other aspects.

* Jayanta (Nyayamanjari)

* contains a long discussion on this problem

* justices the acceptance of sabda as a distinct means of knowledge

* gives serious thought to the philosophy of language

* tries to explain the views of his predecessors and evaluates

* the arguments of prominent scholars

* his account of sabda is as much relevant to grammar, rhetoric, and linguistics as it is to logic

* convincingly proves that words do not exist before their production
* also refutes the sphota theory of grammarians and argues that the relationships between words
and meaning is conventional
* AI
* sabda can be compared to the concept of formal string and language.
* language is a set of strings made from some alphabet
if a fixed alphabet = E, then the language formed by
E is called [E.sup.*]
ifE= {a} then
[E.sup.*] = {e. a. aa. aaa. ...}

The Naya system and the AI system are comparable and contrastable. The Naya system exhibits numerous parallels with AI regarding its management of valid and invalid types of knowledge. The AI utilises various methodologies beyond logic and inference. These methodologies, such as semantic networks, conceptual graphs, frames, rule-based systems, and processes, provide recommendations for effectively representing information. This contrasts with the Naya and Western schools, which excessively emphasise rectifying the identity of the locus of knowledge acquisition. The latter may currently hold little significance for artificial intelligence. Nonetheless, there exists an urgent necessity to pursue further investigation into various Indian systems, including the neural network system and Buddhist logic, to formulate suitable models of human cognition applicable to artificial intelligence.
Nyāya's method of distinguishing valid from invalid information parallels the approach of artificial intelligence in differentiating correct predictions from inaccuracies. False inference, or mithyājñāna, can be likened to algorithmic bias or misclassification in artificial intelligence (AI). The Nyāya methodology rectifies faults by reasoning, akin to the training of artificial intelligence models, where iterative learning diminishes inaccuracies. To verify truth claims, artificial intelligence could employ logical validation akin to Nyāya. Contemporary artificial intelligence mistake-detection frameworks may draw inspiration from Nyaya's categorisation of perceptual and inferential faults utilised in error analysis. Hierarchical knowledge graphs in artificial intelligence exhibit similarities to the structured cognitive method of Nyāya.

Collaborative Reasoning between Humans and Artificial Intelligence: The integration of human logical reasoning (Nyāya) with machine inference (AI) offers the potential to enhance interpretability and accountability. The logical framework of Nyāya emphasises the cause-effect relationship, termed vyāpti, which asserts that a specific cause invariably produces a specific outcome. Smoke indicates the presence of fire. In the realm of artificial intelligence (AI), this parallels the notion of causal reasoning and explainable AI (XAI), which entails guaranteeing that an AI's output is not just statistically valid but also logically and causally justified. This concept is fundamental to Nyāya logic.

In comparing the Nyāya system with contemporary artificial intelligence (AI) systems, it is crucial to recognise that each system adopts a distinct approach to knowledge. The neural network system exhibits notable similarities with artificial intelligence (AI), especially in its systematic distinction between valid (pramā) and invalid (aprāmā) forms of knowledge. The two frameworks recognise that knowledge transcends mere information acquisition; it must possess specific attributes of trustworthiness, accuracy, and justification to be deemed knowledge. Conversely, although the Naya system relies predominantly on structured logical reasoning and accurately defined epistemic categories, modern AI systems employ a broader array of methodologies that extend beyond conventional logic and inference alone.The importance of Indian epistemological frameworks for artificial intelligence research remains undiminished. Conversely, there remains a substantial necessity to connect with Indian philosophical traditions, including the Naya system and Buddhist logic, in a more profound manner to cultivate models of cognition that are more intricate and nuanced. Buddhist theories of inference, momentariness, and dependent origination offer alternative frameworks for elucidating thinking processes. These theories do not assume the presence of a permanent self, aligning with the non-subjective nature of artificial intelligence systems. The Naya system's thorough analysis of error, uncertainty, and validation can yield significant insights for the advancement of artificial intelligence systems that are more robust, transparent, and epistemically responsible.

Knock your brain

1. Nyāya philosophy is primarily concerned with:

a) Liberation through devotion

b) Logical reasoning and valid knowledge

c) Ritual practices

d) Meditation techniques

2In Nyāya, valid knowledge is called:

a) Avidyā

b) Pramā

c) Samsāra

d) Mokṣa

3 How many Pramāṇas are accepted by the Nyāya school?

a) Two

b) Three

c) Four

d) Six

4 Which of the following is NOT a Nyāya Pramāṇa?

a) Perception (Pratyakṣa)

b) Inference (Anumāna)

c) Comparison (Upamāna)

d) Presumption (Arthāpatti)

5 Anumāna (inference) is based on:

a) Direct experience

b) Testimony

c) Universal relation (Vyāpti)

d) Guesswork

Unleash your potential

1.Elucidate the theory of knowledge and pramāṇas within the Nyaya framework?

2.Elucidate the Nyaya theory of inference, providing appropriate instances?